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Learning Forward and Inverse Kinematics Maps Efficiently

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Abstract— **When learning forward and inverse kinematics maps of manipulators, usually little attention is paid to data-efficiency, i.e., the accuracy gained per action-outcome sample.**

This paper examines properties of popular (online) learning techniques and demonstrates that – regardless of the employed exploration strategy – the structure of kinematics mappings does not allow for a *practically viable* trade-off between the number of samples and the resulting approximation error for manipulators with more than a few DoFs – unless *tailored* parametric models are employed.

We discuss suitable choices for these parametric models for both rigid and elastic discretely-actuated robots and compare their data-efficiency to that of popular exploratory learning approaches relying on non-parametric models. Our theoretical considerations are confirmed by various experimental results for inverse kinematics mappings of rigid and omnielastic manipulators.

I. Introduction

For several decades, roboticists have been applying machine learning techniques to obtain forward kinematics mappings (KMs), inverse kinematics mappings (IKMs), and inverse dynamics mappings (IDMs) of robot manipulators although well-established analytic modeling techniques exist – in particular for manipulators with negligible to moderate joint and link elasticity. Online learning methods for (I)KMs and IDMs have become a staple in developmental robotics [1] and nicely fit with the established fundamental principles of incremental development and gradual exploration [2]. However, one crucial aspect for their practical application has been neglected so far: *data-efficiency*, i.e., the accuracy gained per sample.

Exploratory learning techniques for (I)KMs ultimately rely on some (adaptive) sampling strategy in comparatively *high-dimensional* spaces, e.g., the m-dimensional space of generalized joint positions. In contrast to many other application areas, the costs associated with sampling manipulator configurations are non-negligible in terms of *time, tear*, and *wear*. If the dimensionality of the action space is increased, the required number of samples and thus the associated costs typically grow *exponentially*. Even for typical numbers of DoFs m, e.g., $m = 6$, it is therefore necessary to mitigate the effects of the curse of dimensionality (CoD) and minimize the required number of samples to make sampling-based learning approaches practically viable.

One option to reduce the number of samples is to exploit structural properties of the mapping to be learned. However, most approaches for learning (I)KMs employ (local) parametric or non-parametric models which disregard crucial properties of (I)KMs. Regarding (I)KMs of articulated robots, for instance, local linear maps and other local models ignore their most fundamental property: their periodicity.

We show theoretically and experimentally that learning approaches cannot achieve practically viable accuracy with a tolerable number of samples even for manipulators with moderate numbers of joints – unless they *exploit structural properties* of the mapping to be learned. Utilizing a set of problem-tailored regressors, we reformulate the learning problem as a classical linear regression problem (e.g. [3]) and demonstrate the remarkable gain in data-efficiency. Moreover, our results indicate that *inverse* kinematics solutions based on applying algebraic and numerical root-finding techniques to the mentioned parametric forward mappings are superior to directly learned inverse mappings in terms of accuracy and data-efficiency.

Link and joint elasticity are often put forward as arguments to apply sophisticated learning methods to (I)KMs. However, our theoretical and experimental findings prove that parameter estimation approaches employing tailored sinusoidal regressors can capture the forward and inverse kinematics of highly-elastic discretely-actuated robots more efficiently and accurately.

The following section reviews related work in analytic modeling, parameter estimation, and goal-directed learning. Sec. III analyses kinematics mappings of both rigid and elastic manipulators in the Fourier domain. In Sec. IV, the latter results are utilized to devise suitable sets of regressors, formulate corresponding parameter estimation problems, and derive theoretical bounds for the required number of samples of different estimation and learning approaches. Sec. V presents experimental results for the kinematics of a rigid and an omnielastic manipulator underlining our theoretical considerations of the previous sections.

II. Related Work

We briefly review analytic models and regression techniques before sketching popular goal-directed learning approaches for (I)KMs which will serve as basis for comparison.

A. Analytic Models and Parameter Regression

Rigid body manipulators aside, various analytic models based on the (linear) theory of elasticity have been proposed for both continuum (e.g. [4]) and discretely-actuated elastic robots (e.g. [5]–[7]). As various effects and interdependencies between effects are not captured by these models, e.g., shear deformations of the links or nonlinear characteristics

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of the joint elasticities, these analytic models often do not yield satisfactory results. We show that our parametric models surpass the accuracy of analytic models [7] for an omnielastic, articulated robot.

In the context of calibration, usually *small* deviations of a set of kinematics parameters (e.g. DH parameters) in an existing analytical model are considered [8]. As the parameters occur nonlinearly in the model, the model is usually linearized locally to simplify solution (e.g. [5], [9]) starting from initial parameter estimates derived from CAD data, etc. In this paper, however, we are concerned with learning (I)KMs from scratch – without any prior information on the parameters.

Viewed as a more general *direct* modeling problem, various regression techniques can be applied to learn KMs at the position level or the velocity level. Direct model learning approaches using standard regression techniques such as (recursive) least-squares [3] based on polynomial models [10], neural networks (e.g. [11]), support vector machines (e.g. [12]), etc. have found widespread application in kinematics and dynamics model learning. However, none of the established approaches such as LWPR, LGPR, IMLM, ILO-GMR, LoSVR, iRFRLS, and XCSF [13] exploits the *periodicity* of the KM of articulated robots, let alone its *spectral characteristics*.

Considering periodicity and spectral properties, truncated multivariate Fourier series (MFS) (e.g. [14]) are an obvious but largely disregarded choice for regressors. For discretely-actuated robots with negligible joint and link elasticity, a suitable set of regressors can be derived from their transformation matrices but for robots with significant elasticity, it is not immediately apparent how to select an *effective* and *efficient* set of regressors from MFS. In Sec. III we analyze kinematics mappings of rigid and elastic discretely-actuated manipulators and derive suitable choices for regressors based on MFS in Sec. IV.

To apply regression techniques to IKMs, their non-uniqueness has to be handled, e.g., by introducing additional constraints [15]. Alternatively, numeric techniques can be applied to obtain inverse solutions based on the forward model. We opt for a combination of symbolic and numeric techniques as it preserves redundant solutions and still allows for very fast computation.

B. Goal-Directed Learning

Human-inspired movement skill learning and motion control are popular topics in both neuroscience and machine learning which aim at describing and replicating human development and learning processes. A broad variety of learning schemes has been proposed to imitate and replicate human motor learning and control based on internal models [16]. In particular, various approaches to learn (I)KMs offline or online from scratch can be found in the literature.

Established approaches for learning *inverse* models are distal teachers [17] and feedback error learning [18]. Learning *forward* models by motor babbling was proposed in [19]. In motor babbling, training samples are acquired by *random* exploration of motor commands and the action-outcome mapping is learned.

Recent studies, however, proved that even infants do not explore their "workspace" randomly as implied by motor babbling; instead, they execute goal-directed motion [20] already few days after birth. Such intrinsically motivated goal-directed learning approaches to obtain IKMs following the "learning while behaving" paradigm were proposed in [21], [22] and the term Goal Babbling was coined. Apart from Goal Babbling, various other schemes for direct [23] and incremental [24] learning of inverse models were proposed.

As Goal Babbling requires a predefined set of targets and only provides locally valid IKM solutions, direction sampling [25] was introduced as an extension of Goal Babbling in order to explore the *entire* workspace of a manipulator and learn the IKM simultaneously.

Various exploration and learning strategies have been investigated (e.g. [21], [26]–[28]) but only few aimed at reducing the number of required samples (e.g. [29]).

Due to the employed *local* parametric or non-parametric models as well as the online learning and exploration strategies, high numbers of samples are commonly encountered in most goal-directed (I)KM learning approaches.

III. Kinematics in the Fourier Domain

The kinematic forward transformation of a rigid or elastic discretely-actuated serial manipulator can be shown to be well-behaved and in particular C^∞-continuous. The following subsections examine the properties of forward kinematics mappings in the Fourier domain. First, rigid manipulators are considered. Subsequently, link and joint elasticity are incorporated into the analysis. Finally, limitations of the approach are addressed.

A. Rigid Body Kinematics

Let the forward kinematics transformation of a serial manipulator with m joints be given by the homogeneous transformation matrix ${}^{\boldsymbol{B}}_{\boldsymbol{E}}\boldsymbol{T}$, where $\boldsymbol{B}$ denotes the base frame of the manipulator and $\boldsymbol{E}$ the end-effector frame. Let $\mathcal{R}$ and $\mathcal{P}$ denote the sets of joint variables associated with revolute and prismatic joints resp. The matrix elements ${}^{B}_{E}t_{uv}, u \in \{1,2,3\}, v \in \{1,\dots,4\}$ of ${}^{\boldsymbol{B}}_{\boldsymbol{E}}\boldsymbol{T}$ are functions of the joint variables $q_1,\dots,q_m$, i.e. ${}^{B}_{E}t_{uv} = f(q_1,\dots,q_m)$. They are 2π-periodic functions for $q_i = \theta_i$ corresponding to revolute joints and linear in the joint variables $q_j = d_j$ associated with prismatic joints. The scripts B and E of ${}^{B}_{E}t_{uv}$ will be omitted from now on.

Transforming the elements t_{uv} into the Fourier domain, one obtains Dirac delta distributions (DDD) originating from revolute joints $\theta_j \in \mathcal{R}$ and first derivatives of DDDs originating from prismatic joints $d_j \in \mathcal{P}$. First, manipulators with only revolute joints are considered, i.e., $\mathcal{P} = \emptyset$. Second, the analysis is extended to manipulators with both revolute and prismatic joints.

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1) Rigid Serial Manipulators with Revolute Joints Only: Any element t_{uv} of ${}^{\boldsymbol{B}}_{\boldsymbol{E}}\boldsymbol{T}$ can be represented as

$$t_{uv}(\boldsymbol{\theta}) = \sum_{i \in \mathcal{I}_m^1} p_i \prod_{j=1..m} f_{i_j}(\theta_j) \tag{1}$$

where

$$f_{i_j}(\theta_j) = \begin{cases} 1 & \text{if } i_j = 0 \\ \cos(\theta_j) & \text{if } i_j = 1 \\ \sin(\theta_j) & \text{if } i_j = -1 \end{cases} \tag{2}$$

$p_i \in \mathbb{R}$, $\mathcal{J}_m^1 = \{(j_1, \ldots, j_m) | j_k \in \{-1, 0, 1\}, k \in \{1, \ldots, m\}\}$ is a set of m-tuples denoting indices and $\mathcal{I}_m^1 \subseteq \mathcal{J}_m^1$. Note that the total number of m-tuples and hence the maximum number of coefficients p_i in Eq. (1) is bounded by $n_t = 3^m$.

Computing the multivariate Fourier transform (FT) of t_{uv} yields

$$T_{uv}(\boldsymbol{\Theta}) = \sum_{i \in \mathcal{I}_m^1} p_i \prod_{j=1..m} F_{i_j}(\Theta_j) \tag{3}$$

where $FT(f_{i_j}(\theta)) = F_{i_j}(\Theta)$, i.e.,

$$F_{i_j}(\Theta) = \begin{cases} \delta(\Theta) & \text{if } i_j = 0 \\ \pi(\delta(\Theta+1) + \delta(\Theta-1)) & \text{if } i_j = 1 \\ \iota\pi(\delta(\Theta+1) - \delta(\Theta-1)) & \text{if } i_j = -1 \end{cases} \tag{4}$$

Note that ι denotes the imaginary unit here and δ represents the DDD.

All terms of T_{uv} are bounded by an m-hypercube of edge length 2 in the Fourier domain. In particular, the vertices of the hypercube are given by the 2^m m-tuples $(\pm 1, \ldots, \pm 1)$. Therefore, the highest "frequencies" or rather repetencies will occur along the diagonals of the hypercube and are given by

$$\Theta_{max} = \sqrt{m} \tag{5}$$

Fig. 1 illustrates this property by depicting the FT of the x-coordinate of the kinematics map of a 2R and a 3R manipulator. The diamonds represent DDDs with different coefficients.

2) Manipulators with both Prismatic and Revolute Joints: If prismatic joints are present, each prismatic joint is associated with a joint variable d_j which occurs linearly in ${}^{\boldsymbol{B}}_{\boldsymbol{E}}\boldsymbol{T}$. Thus, f_{i_j} for prismatic joints is given by

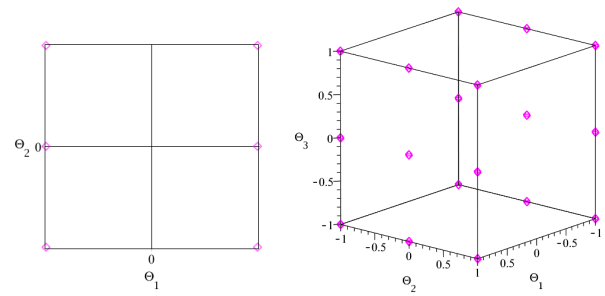
$$f_{i_j}(d_j) = \begin{cases} 1 & \text{if } i_j = 0 \\ d_j & \text{if } i_j = 1 \end{cases} \tag{6}$$

Note that $i_j = -1$ does not occur for prismatic joints.

A linear function corresponds to the first derivative of a DDD in the Fourier domain.

$$FT(d_j) = \frac{d\delta(D_j)}{dD_j} \tag{7}$$

In the following analysis, manipulators with revolute joints only are considered as the presence of prismatic joints does neither increase the complexity of T_{uv} nor that of the index set $\mathcal{I}_m^1$.



(a) (b)

Fig. 1: Exemplary FTs of the x-coordinates of the kinematics maps of a 2R manipulator (a) and a 3R manipulator (b). DDDs are visualized as diamonds. Note that they are bounded by a square (2R) and a cube (3R) resp. with an edge length of 2. Hence, the maximum repetencies occurring along the diagonals are $\sqrt{2}$ (2R) and $\sqrt{3}$ (3R).

B. Link Elasticity

The previous analysis did not consider link and joint elasticity. If link elasticity cannot be neglected, the resulting deformations generate contributions in the Fourier domain which are not restricted by the mentioned m-hypercube any longer but contain terms with higher repetencies. Depending on the geometry and elastic properties of the link as well as the kinematic structure of the manipulator, various significant terms with repetencies higher than 1 may occur in $T_{uv}(\boldsymbol{\Theta})$.

Link elasticity can be considered in the KM by incorporating link deformations according to the (linear) theory of elasticity. As this paper is concerned with the *qualitative* properties of the resulting kinematics map, Euler-Bernoulli beam theory instead of more sophisticated approaches is used. Hence, shear deformations of the links and nonlinearities are neglected; only bending and tension/compression of the links are considered. To further simplify analysis, bending and tension/compression deformations are assumed to be independent and moreover, the deformations are assumed to have negligible effects on the distribution of masses along the kinematic chain.

By considering the superimposed effects of bending and compression deformation on a transformation between two rigid links of a manipulator according to the DH convention, the following equation results:

$${}^{i-1}\mathfrak{A}_i = {}^{i-1}\boldsymbol{A}_i \; {}^{i}\boldsymbol{T}_{bc} \; {}^{i}\boldsymbol{R}_b \tag{8}$$

${}^{i}\boldsymbol{T}_{bc}$ and ${}^{i}\boldsymbol{R}_b$ are assumed to be small translations and rotations resulting from compression and bending and are therefore considered as *differential* transformations.

While each link transformation matrix ${}^{i-1}\boldsymbol{A}_i = {}^{i-1}\boldsymbol{A}_i(\theta_i)$ is a function of θ_i only, ${}^{i}\boldsymbol{T}_{bc}$ and ${}^{i}\boldsymbol{R}_b$ already contain products of like sine and cosine terms which can be inferred from the kinematics equations derived in [5], [6] based on Timoshenko and Euler-Bernoulli beam theory resp.

Hence, ${}^{i-1}\mathfrak{A}_i$ contains products of sines and cosines $\cos^{k_c}(\theta_i)\sin^{k_s}(\theta_j)$, $k_s, k_c \in \mathbb{N}$, $i, j \in \{1, \ldots m\}$. There-

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fore, the forward kinematics map given by the product

$$ {}^{\mathbf{B}}_{\mathbf{E}}\mathfrak{T} = \prod_{i=1}^{m} {}^{i-1}\mathfrak{A}_i \tag{9} $$

may contain products of even higher order.

sin and cos will be abbreviated by s and c for the remainder of the analysis. Let $a_\bullet \in \mathbb{R}$ denote coefficients. Using the power reduction formulae for trigonometric functions, for instance,

$$ c^{k_c}(\theta_i)s^{k_s}(\theta_j) = C_{k_c}(\theta_i) \cdot \begin{cases} C_{k_s}(\theta_j) & k_s \text{ even} \\ S_{k_s}(\theta_j) & k_s \text{ odd} \end{cases} \tag{10} $$

$$ C_k(\theta) = \sum_{v=0}^{k} a_{v,k} c(v\theta) \tag{11} $$

$$ S_k(\theta) = \sum_{w=1}^{k} a_{w,k} s(w\theta) \tag{12} $$

and the product-to-sum formulae for products $s(k_1\theta)s(k_2\theta)$, $s(k_1\theta)c(k_2\theta)$, etc. ($k_1, k_2 \in \mathbb{N}$), e.g. $s(k_1\theta)s(k_2\theta) = \frac{1}{2}\left(c((k_1-k_2)\theta) - c((k_1+k_2)\theta)\right)$, any term in Eq. (9) can be represented by a product or a sum of products of trigonometric functions with repetencies ≥ 1. Note, however, that the maximum repetency is limited by $k_c + k_s$ and the coefficients decline rapidly with increasing repetency. For instance, when expanding $\sin^5(\theta)\cos^5(\theta)$, the coefficient of $\sin(10\theta)$ is only one tenth of that of $\sin(2\theta)$.

C. Joint Elasticity

Assuming that the elasticity of the distal joints will not significantly affect the distribution of masses in the partial kinematic chain $q_{i+1}, \ldots, q_m$, joint elasticity can be modeled by

$$ {}^{i-1}\mathfrak{A}_i = {}^{i-1}\boldsymbol{A}_i(q_i + \Delta_i) \tag{13} $$

where $\Delta_i = c_i g_i^{i+1,m}(q_{i+1}, \ldots, q_m)$, c_i is the compliance of joint i and $g_i^{i+1,m}(q_{i+1}, \ldots, q_m)$ is the generalized gravitational force depending on the configuration of joints $i+1, \ldots, m$ acting along/about the z axis of joint i. As for small Δ_i, $\sin(q_i + \Delta_i) \approx \sin(q_i) + \cos(q_i)\Delta_i$ and $\cos(q_i + \Delta_i) \approx \cos(q_i) - \sin(q_i)\Delta_i$ holds, and $\Delta_i = c_i g_i^{i+1,m}(q_{i+1}, \ldots, q_m)$ is a trigonometric polynomial, ${}^{i-1}\mathfrak{A}_i = {}^{i-1}\mathfrak{A}_i(q_i, \ldots, q_m)$ can be approximated by a trigonometric polynomial as well. When computing the forward kinematics map according to Eq. (9), higher powers of sines and cosines result again. By applying Eq. (10)-(12), these higher powers can again be reduced to terms of higher repetencies.

Therefore, regardless of the type of elasticity encountered, the Fourier transform of a kinematics map of an elastic manipulator with only revolute joints exhibits non-negligible terms with repetencies ≥ 1 but still all t_{uv} remain periodic functions. Moreover, the repetencies are limited and the coefficients of terms with higher repetency tend to be small. These properties can be exploited to tailor appropriate parametric models with a reduced number of parameters.

D. Limitations

The previous analysis applies to discrete-joint manipulators and cannot readily be extended to most continuum manipulators as their kinematics mapping does not show any periodicity.

IV. Models for Parameter Estimation

This sections considers three different models of kinematics mappings: the parametric model based on the well-known DH parameters, which is usually employed in calibration approaches; models based on multivariate truncated Fourier series, which are natural but neglected choices for parametric models; and models based on (smoothed) piecewise linear functions which constitute the basis of various learning approaches. Finally, their required numbers of samples to reach a desired accuracy are compared.

A. Estimating Denavit-Hartenberg Parameters

The forward kinematic transformation of discretely-actuated serial manipulators with negligible joint and link elasticity can be represented by a minimal set of three parameters and one joint variable (potentially with an offset) per prismatic or revolute joint following the well-established DH convention or any of its variants or generalizations [30]–[32]. Following these conventions, the number of parameters to represent the kinematics map of a serial manipulator with m joint DoFs is $n_{DH} = 4m$ including offsets in the joint variable.

These parameters occur nonlinearly in the kinematics map, i.e., as products of (trigonometric functions of) parameters. The resulting estimation problem is therefore nonlinear if the DH parameters are to be identified directly. Classical approaches in robot calibration often use first-order approximations thus linearizing the estimation problem about the expected parameter values. Details on their identifiability can be found in [33].

B. Multivariate Fourier Series (MFS)

In Sec. III, we demonstrated that even if joint and link elasticity are considered, the kinematics map of a manipulator remains a periodic function. In contrast to the case of rigid links, terms with repetencies $\Theta > 1$ occur in the elastic case. MFS therefore appear to be a natural and efficient choice to represent kinematics maps of both rigid and elastic manipulators.

1) Rigid Manipulators: In contrast to the direct estimation of DH parameters, estimating the coefficients of an MFS is a linear problem. For rigid manipulators, Eqs. (1), (2), and (6) can be utilized directly to estimate the parameter vector $\boldsymbol{p}$. Let $n_{pr} = |\mathcal{P}|$ and $n_{rev} = |\mathcal{R}|$ denote the numbers of prismatic and revolute joints, respectively. For the *full* index set, i.e., $\mathcal{J}_m^1$ in Eq. (1), the number of parameters to be determined *for each* t_{uv} is given by

$$ n_{MF,\mathcal{I}_m^1} = 2^{n_{pr}} 3^{n_{rev}} \tag{14} $$

However, in the rigid case $\mathcal{I}_m^{\min} = \mathcal{I}_m^1 \subset \mathcal{J}_m^1$ holds and a large fraction of the parameters are (ideally) zero.

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TABLE I: Number of parameters depending on the selected index set and the number of joints

Number of joints m	2	3	4	5	6	7	8
Min. no. of params.	2	6	18	54	162	486	1458
Max. no. of params.	6	18	54	162	486	1458	4374
Cardinality of $\mathcal{J}_m^1$	9	27	81	243	729	2187	6561

When employing minimum index sets $\mathcal{I}_m^{\min} = \mathcal{I}_m^1$ instead of the full index set $\mathcal{J}_m^1$ for the t_{uv}, the number of parameters p_i (cf. Eq. 1) in each t_{uv} varies. Tab. I gives the min. and max. number of parameters p_i in the elements t_{uv} for different numbers of revolute joints as well as the total number of tuples in the full index set $\mathcal{J}_m^1$. Obviously, the min and the max. number of parameters are only $\frac{2}{9}$ and $\frac{2}{3}$ resp. of the cardinality $|\mathcal{J}_m^1|$ of the full index set. If offsets in the joint variable are assumed to be known, the number of parameters decreases further, e.g., for $m = 8$, the min. and max. no. of params. reduce to 1364 and 377 resp.

2) Elastic Manipulators: Sec. III suggests to consider sinusoidal terms of higher repetency, i.e., $\Theta > 1$, in order to consider joint and link elasticity in kinematics models. If repetencies of up to Θ_{max} are considered, i.e., $i_j \in \{-\Theta_{max}, \ldots, \Theta_{max}\}$, the functions f_{i_j} for revolute joints (cf. Eq. (2)) change to

$$f_{i_j}(\theta_j) = \begin{cases} 1 & \text{if } i_j = 0 \\ \cos(i_j\theta_j) & \text{if } i_j > 0 \\ \sin(i_j\theta_j) & \text{if } i_j < 0 \end{cases} \tag{15}$$

and the number of parameters increases to $n_{MF,\mathcal{J}_m^{\Theta_{max}}} = (2\Theta_{max} + 1)^m$

By substituting Eq. (15) into Eq. (1), an MFS results again. The number of its parameters grows exponentially with the number of DoFs.

3) Hyperbolic Crosses: To mitigate the curse of dimensionality, hyperbolic crosses (HCs) are employed in numerical approximation [34], [35]. Instead of considering the full index set $\mathcal{J}_m^{\Theta_{max}}$ with repetencies of up to Θ_{max} in m variables, only the subset

$$\mathcal{H}_m^{b,\Theta} = \left\{ i = (i_1, \ldots, i_m) \in \mathbb{Z}^m : |i_j| \leq \Theta \wedge \prod_{j=1}^{m} \max(|i_j|, 1) \leq b \right\} \tag{16}$$

is considered. This choice of indices reduces the number of parameters from $n_{MF,\mathcal{J}_m^{\Theta_{max}}} = (2\Theta_{max} + 1)^m$ to $n_{MF,\mathcal{H}_m^{b,b}} = c_m b \left(\ln(b)\right)^{m-1}$ where c_m is a dimension-dependent constant [36], [37].

Instead of simple (random) sampling strategies, lattice rule algorithms for hyperbolic crosses from the field of numerical approximation (e.g. [38], [39]) can be adopted to permit optimal selection of samples in terms of data-efficiency.

4) Estimation Problem: To determine the kinematic forward transformation of a serial manipulator with revolute joints based on sampling joint configurations and observing the end-effector poses, Eqs. (2) or (15) and (1) are utilized. As the latter is linear in the parameters p_i, an overdetermined system of linear equations can be easily obtained from n_s sampled joint configurations and end-effector frames $\boldsymbol{E}$. Neglecting structural and statistical dependencies among the t_{uv} as well as constraints arising from $\boldsymbol{E} \in SE(2)$ or $\boldsymbol{E} \in SE(3)$ resp., the regression problem for each t_{uv} can be stated as

$$\boldsymbol{V}(\boldsymbol{\theta})\boldsymbol{p} = \boldsymbol{w} \tag{17}$$

where $\boldsymbol{V}(\boldsymbol{\theta})$ is an $n_s \times n_p$ matrix depending on the joint variables $\boldsymbol{\theta}$, $\boldsymbol{p}$ denotes the n_p-vector of parameters (coefficients) and $\boldsymbol{w}$ denotes the n_s-vector of observations; e.g., the vector of measured x-coordinates of the end-effector in the case of t_{14}.

The k-th row of $\boldsymbol{V}(\boldsymbol{\theta})$ corresponds to the k-th sample and $v_{k,i} = \prod_{j=1..m} f_{i_j}(\theta_j)$ where the m-tuple i corresponds to the coefficient p_i. Any ordinary or recursive (total) least-squares approach may be applied to compute the parameter estimates.

5) Symbolic-Numeric Inverse Solution: Instead of directly estimating the IKM, symbolic algebraic and numerical root-finding techniques are applied to the KM to obtain inverse kinematics solutions. Basically, the optimization problem

$$\underset{\substack{\mathbf{q} \\ c(\mathbf{q})<0}}{\operatorname{argmin}} \|\boldsymbol{w} - \boldsymbol{f}_{fwd}(\boldsymbol{q}))\| \tag{18}$$

where $\boldsymbol{f}_{fwd}$ denotes the forward kinematics mapping and $\|\cdot\|$ is a suitable norm (or generalized norm if $\boldsymbol{w}$ is non-commensurate), can be simplified computationally by applying algebraic decomposition techniques. More precisely, the obtained MFS for the elements t_{uv} are converted to polynomials using a suitable substitution, e.g., tangent half-angle substitution. Subsequently, a (partial) Gröbner basis (e.g. [40]) or triangular decomposition (e.g. [41]) w.r.t. a selected order of variables is computed to simplify numerical solution. The roots of the resulting system of polynomials and finally the joint variables are determined numerically.

C. Local Linear Models

In Sec. III-A, the maximum repetency $\Theta_{max} = \sqrt{m}$ of any kinematics map of a manipulator without joint or link elasticity has been derived. Assuming symmetric and identical joint limits q_{lim}, a distance

$$s = \sqrt{\sum_{i=1}^{m} (2q_{lim})^2} = 2q_{lim}\sqrt{m} \tag{19}$$

along the diagonals in joint space has to be covered by sampling. Along those diagonals, a maximum repetency of Θ_{max} can occur and hence $\frac{\Theta_{max}s}{2\pi}$ periods have to be covered if tuples $i = (\pm 1, \ldots, \pm 1) \in \mathcal{I}_m^1$ occur in t_{uv}. Assuming $q_{lim} = \pi$, $\frac{\Theta_{max}s}{2\pi} = m$ periods have to be covered.

The maximum relative error of approximating a sinusoidal function $a\cos(\theta + \phi)$, where $\phi, a \in \mathbb{R}$, with a piecewise linear function can easily be shown to be

$$e_{rel} = 1 - \cos\left(\frac{\pi}{n_{spp}}\right) \tag{20}$$

where n_{spp} is the number of samples *per period*. Assuming an *equidistant* distribution of samples, a piecewise *linear*

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function requires $n_{spp} = 23$ to achieve a relative error of 1% and $n_{spp} = 32$ for an error of 0.5%.

As the kinematic structure of the manipulator is assumed to be unknown, $n_{spp}m$ (equidistant) samples have to be generated along each diagonal to achieve the desired error. The worst-case number of samples thus roughly amounts to

$$n_{LL} = n_{spp}^m m^m \tag{21}$$

which is orders of magnitude greater than n_{MF} from the preceding subsections. Note that sampling schemes which adapt the sample density to the observed mapping error as well as higher-order (polynomial) or Gaussian models, etc. can reduce the required number of samples.

D. Comparison of the Number of Required Samples

Fig. 2 shows the theoretical minimum number of samples required to learn the kinematics mapping of an m DoF manipulator using different models: DH parameters (DH), MFS with $\Theta_{max} = 1$ (SC1) to $\Theta_{max} = 3$ (SC3), MFS with an index set according to the hyperbolic cross $\mathcal{H}_m^{4,3}$ (HC4), and piecewise linear models with maximum permissible errors of 1% (LL1) and 0.5% (LL05). The number of DH

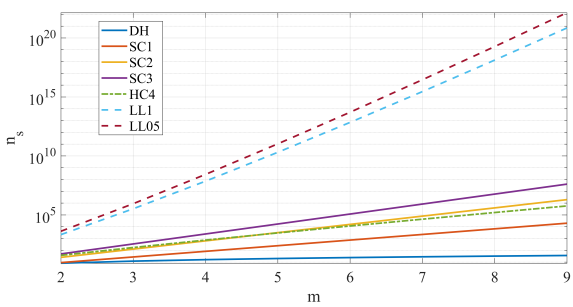


Fig. 2: Minimum number of samples n_s required for several parametric models to learn the KM of a manipulator depending on its number of joints m.

parameters grows linearly with m. Although the required number of samples grows exponentially for SC1-SC3, it still remains manageable in real-world scenarios. Index sets according to hyperbolic crosses are particulary favorable as they significantly reduce the number of parameters. In fact, HC4 requires less samples than SC2 for manipulators with a higher number of joint DoFs. In contrast, the required number of samples for piecewise linear models (LL1, LL05) quickly becomes intractable. Note that the number of samples must not be confused with the number of iterations/epochs in GB or other learning approaches. The former is usually by orders of magnitude higher than the latter.

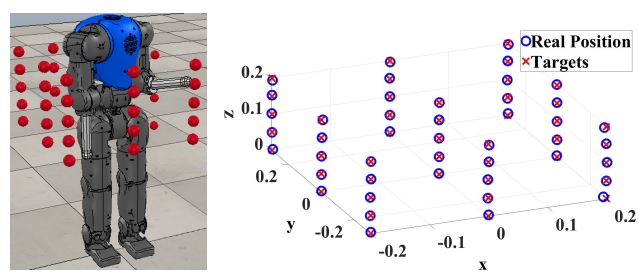
V. Experimental Results

This section presents several experimental results for goal-directed learning and regression approaches illustrating and validating the above claims. First, learning the inverse kinematics of a manipulator with no joint and link elasticity is considered. Secondly, the inverse kinematics map of an omnielastic manipulator is addressed.

A. Inverse Kinematics of a Rigid Manipulator

In this experiment, learning the inverse kinematics of a 9-DoF COmpliant huMANoid (COMAN) floating-base robot (6 DoF floating base + 3DoF arm) without any prior parameter knowledge was considered. The root mean square error (RMSE) of a popular goal-directed learning approach, viz. Goal Babbling (GB) using local linear maps (LL) as incremental regression mechanism, henceforth called GBLL, was compared to the RMSE obtained by the proposed least-squares regression approach relying on MFS and the minimum index set $\mathcal{I}_m^{min}$ for rigid manipulators. Note that the model obtained by GB provides a *single* joint configuration for each pose whereas the symbolic-numeric inversion in MFS permits the computation of all solutions.

White Gaussian noise (iid) with $\sigma \in \{1mm, 2mm, 3mm\}$ was added to all sampled end-effector positions to model measurement noise. As GB is based on reaching a set of specified targets, 45 targets were distributed on a regular 3D grid as illustrated in Fig. 3(a): 15 targets were in front of the robot, 15 behind the robot, and 15 in the coronal plane (i.e., the robot needed to *walk* to reach these targets). Subsequently, the IKM was learned for these targets using GBLL. Fig. 3(b) shows that the robot managed to reach all targets quite accurately (after $100,000$ training samples). The training and test target set were identical for GBLL while for MFS, separate training and test sets were created by random sampling in configuration space following an iid uniform distribution. The cardinality of the test set in MFS was $100,000$ samples. Fig. 4 shows the RMSEs obtained



(a) (b)

Fig. 3: (a) COMAN with targets indicated by red spheres; (b) targets and reached positions.

for various numbers of samples and σ. For each number of samples, 16 trials of GBLL and 100 trials of MFS have been performed. The standard deviation of the results is indicated by the shaded areas. MFS yield more accurate inverse kinematics solutions with less samples than GBLL. While the performance of MFS is clearly superior, GBLL proves to be less affected by measurement noise.

B. Inverse Kinematics of an Omnielastic Manipulator

The inverse kinematics mapping of the omnielastic TUDOR robot was estimated based on the MERIt data set [42]. The performance of several variants of MFS differing w.r.t. the employed regressors, direct estimation of DH parameters, and a function fitting neural network (FN) (Matlab `fitnet`

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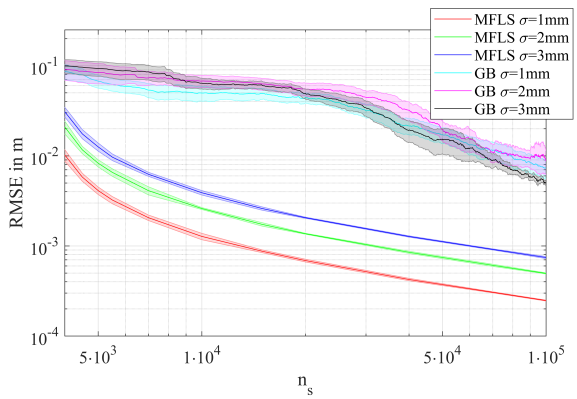

Fig. 4: RMSE of MFS and GBLL over number of samples for different standard deviations σ of the measurement noise. Note that the shaded areas indicate $0.3s$ for GBLL and $3s$ for MFS where s denotes the sample standard deviation.

command) were compared to each other. In each case, the *direct* KM has been learned, and the mentioned symbolic-numeric inversion technique has been applied to obtain the inverse solutions.

Separate test and training sets have been selected randomly from the available samples. The RMSE is computed using 100 random draws from a given number of samples. For MFS, ordinary least-squares was used.

Fig. 5 shows the RMSE over the number of training samples n_s. As the standard deviation tends to exhibit very high values before convergence, regardless of the employed model, it has been omitted in the plot. The error of the DH

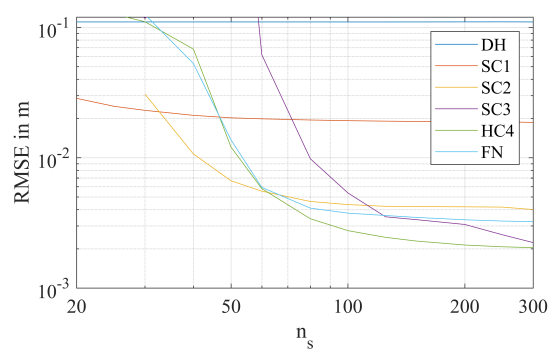

Fig. 5: RMSE of MFS with different regressor sets and FN over the number of training samples n_s for the inverse kinematics mapping of the omnielastic robot TUDOR. DH indicates the kinematics model based on the DH convention. SC1, SC2, and SC3 denote MFS with maximum repetencies of 1, 2, and 3 respectively; HC4 indicates SC3 ($\Theta_{max} = 3$) with a reduced index set according to the hyperbolic cross $\mathcal{H}_3^{4,3}$. FN denotes the result obtained by training a function fitting neural network for the direct KM (`fitnet`).

model already converges for as little as $n_s = 10$ training samples but the remaining error of this rigid-body kinematics model is very high due to the elasticity of the robot. The error of MFS with a maximum repetency of $\Theta_{max} = 1$ (SC1) also converges already for low numbers of samples but also shows a mean error of about $2cm$ while SC2 ($\Theta_{max} = 2$) can capture the elasticity of the robot much better. The function fitting neural network (FN) requires more samples to stabilize but shows lower errors than SC2. Due to its higher number of parameters, SC3 ($\Theta_{max} = 3$) needs even more samples to converge and yields smaller errors than both SC2 and FN. The accuracy achieved by SC3 and the hyperbolic cross HC4 clearly surpass those of all models evaluated in [7], [43] for the same data set. Owing to its comparatively low number of parameters, HC4 ($\mathcal{H}_3^{4,3}$) offers an ideal compromise between fast convergence and high accuracy.

Fig. 6 illustrates the concept and benefits of hyperbolic crosses. Fig. 6(a) shows the hyperbolic crosses $\mathcal{H}_2^{5,5}$ and $\mathcal{H}_2^{4,3}$(shaded area) for general repetency variables Φ and Ψ. Red crosses correspond to high-repetency terms which are not included in the hyperbolic cross $\mathcal{H}_2^{4,5}$ but in the full index set $\mathcal{I}_2^5$. Likewise, crosses in the shaded area correspond to terms not contained in $\mathcal{H}_2^{4,3}$ but in $\mathcal{I}_2^3$. Fig. 6(b) depicts the normalized amplitude spectrum of the y-coordinate of the TUDOR kinematics map. It shows that high-repetency terms indicated by red crosses (not contained in $\mathcal{H}_2^{4,3}$ but in $\mathcal{I}_2^3$) show very low amplitudes. Their contribution to the kinematics mapping is hence negligible and they can safely be omitted. Therefore, using the index set $\mathcal{H}_2^{4,3}$ instead of $\mathcal{I}_2^3$ provides an efficient way to reduce the number of terms and still maintain high accuracy. Note that in higher dimensions and for higher maximum repetencies, the effect of the hyperbolic cross is even more pronounced.

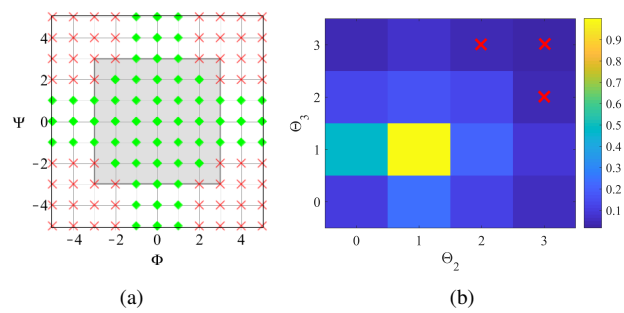

Fig. 6: (a) Visualization of $\mathcal{H}_2^{5,5}$ (green diamonds) and $\mathcal{H}_2^{4,3}$ (green diamonds in shaded area) ; (b) Amplitude spectrum of the y-coord. of the TUDOR fwd. kinematics map (θ_2 and θ_3). Terms not included in $\mathcal{H}_2^{4,3}$ are indicated by red crosses.

VI. Conclusion

We demonstrated that forward and inverse kinematics mappings of discretely-actuated manipulators can be learned effectively and efficiently by estimating the coefficients of suitably selected sinusoidal functions.

In contrast to many popular regression and goal-directed learning approaches, the employed parameter estimation technique is less affected by the curse of dimensionality (CoD). In particular, the CoD can effectively be mitigated

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by (i) selecting sinusoidal regressors which respect the periodicity, spectral characteristics, and structure of the mapping to be estimated; (ii) reducing the number of parameters by selecting regressors according to hyperbolic crosses. These practices were shown to drastically increase data-efficiency, that is, the accuracy gained per sample.

Joint and link elasticity were shown to cause terms with repetencies $\Theta > 1$ in kinematics mappings – hence preserving their periodicity. With a suitably selected set of sinusoidal regressors, the proposed parameter estimation approach was able to achieve remarkable accuracies with low numbers of samples for elastic manipulators as well.

Instead of directly learning an inverse kinematics mapping, we used symbolic-numeric techniques to compute inverse solutions based on the estimated forward mapping. This approach achieved highly accurate results and moreover allows to fully exploit the redundancy of the manipulator.

Our experimental results for both rigid and elastic manipulators clearly show that our proposed method conspicuously outperforms the examined goal-directed learning approaches in terms of data-efficiency and terminal accuracy.

From a practical point of view, regression techniques with *tailored* sets of regressors and symbolic-numeric inversion methods are thus clearly preferable over goal-directed learning approaches for obtaining (inverse) kinematics mappings.

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